

Closed-QA Model Evaluation Report

Databricks Dolly 15k · GPT-4o-mini · LangSmith Experiment

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1. Executive Summary

This report documents the evaluation of GPT-4o-mini on a closed question-answering (closed-QA) task using the LangSmith experimentation platform. The model was assessed against 20 curated examples from the Databricks Dolly 15k dataset, with performance measured across two dimensions: correctness (does the answer match the reference?) and faithfulness (does the answer stay within the supplied context?).

The evaluation pipeline was built end-to-end in Python, incorporating LangSmith tracing, an LLM-as-judge evaluation framework, and automated results analysis. All experiment runs are reproducible and version-tracked in LangSmith (EU region).

Key Findings

- Correctness mean score: 0.83 out of 1.0, with 90% of examples passing the ≥ 0.5 threshold.
- Faithfulness mean score: 0.90 out of 1.0, with 90% of examples fully grounded in context (score = 1.0).
- 75% of examples were fully correct AND fully faithful — a strong baseline result for a zero-shot prompt.
- Primary failure mode: hallucination (10% of examples) — the model introduced claims absent from the context passage.
- Secondary failure mode: incomplete multi-part answers (15%) — the model captured only part of a required response.

2. Methodology

2.1 Dataset

Property	Value
Source	databricks/databricks-dolly-15k (Hugging Face)
Category filter	closed_qa
Examples (post-QA)	20 (quality-filtered from 300+ closed_qa records)
Quality gates	Non-empty question, non-empty answer, non-empty context passage

Example structure	inputs: {question, context} outputs: {answer} metadata: {category, source}
LangSmith dataset	"Dolly Databricks – Closed QA Evaluation Set"

Closed-QA is a constrained task in which the model must derive its answer exclusively from a supplied context passage. This design makes faithfulness a meaningful and independently measurable constraint. Examples span diverse domains including history, science, geography, and general knowledge.

2.2 Target Model & Configuration

Parameter	Value
Model	gpt-4o-mini (OpenAI)
Temperature	0 (deterministic; ensures reproducibility)
Max tokens	256
System prompt	"Answer using ONLY the information in the context. Reply 'I don't know' if the answer cannot be found."
Tracing	LangSmith @traceable decorator, EU endpoint (eu.api.smith.langchain.com)
Experiment prefix	gpt-4o-mini-closed-qa

2.3 Evaluation Approach

Two LLM-as-judge evaluators were implemented, each calling GPT-4o-mini at temperature 0 with a structured JSON response format to eliminate parsing ambiguity.

Correctness Evaluator

- Compares the model's answer against the reference (ground-truth) answer from the dataset.
- Three-point scale: 1.0 = fully correct, 0.5 = partially correct, 0.0 = incorrect or missing.
- Pass threshold: score ≥ 0.5 .

Faithfulness Evaluator

- Assesses whether every claim in the model's answer is explicitly supported by the supplied context passage.
- Binary scale: 1.0 = fully grounded in context, 0.0 = contains at least one unsupported claim.

- Pass threshold: score = 1.0.

All evaluator outputs, model traces, and scores were automatically stored in LangSmith under a single named experiment run, enabling full auditability and one-click replay.

3. Results

3.1 Aggregate Metrics

Metric	n	Mean	Median	Stdev	Min	Max
Correctness	20	0.83	1.00	0.26	0.0	1.0
Faithfulness	20	0.90	1.00	0.31	0.0	1.0

3.2 Score Distribution & Pass Rates

Metric	Score 0.0	Score 0.5	Score 1.0	Pass Rate	Threshold
Correctness	2 (10%)	3 (15%)	15 (75%)	90% (18/20)	≥ 0.5
Faithfulness	2 (10%)	N/A	18 (90%)	90% (18/20)	$= 1.0$

3.3 Overall Performance Summary

Category	Count	Percentage
Fully correct AND faithful (correctness = 1.0, faithfulness = 1.0)	15	75%
Partially correct, faithful ($0.5 \leq \text{correctness} < 1.0$, faithfulness = 1.0)	3	15%
Faithful but wrong (correctness < 0.5, faithfulness = 1.0)	1	5%
Hallucinated — model introduced unsupported claims (faithfulness = 0.0)	2	10%

4. Analysis

4.1 Performance Patterns

GPT-4o-mini performed strongly on straightforward factual retrieval tasks where the answer could be lifted directly from a single sentence in the context. The 75% fully-correct-and-faithful rate demonstrates that a simple zero-shot system prompt is a capable baseline for this task class.

Correctness failures clustered around two recognisable patterns:

- Multi-part questions: when a reference answer contained two or more distinct facts, the model frequently captured only one, yielding a partial score of 0.5.

- Indirect inference: questions requiring synthesis across multiple sentences produced incomplete or imprecise answers.

4.2 Failure Mode: Hallucination (10% of examples)

Two examples received a faithfulness score of 0.0, meaning the model introduced at least one claim not present in the context passage. In both cases the hallucinated content was plausible but factually unsupported — a pattern typical of instruction-tuned models that fall back on world knowledge when context is locally ambiguous.

Field	Content
Question	In what city and year were the first modern Olympic Games held?
Context (excerpt)	The Olympic Games were revived in 1896 in Athens, Greece, at the initiative of Pierre de Coubertin...
Reference answer	Athens, 1896.
Model answer	The first modern Olympic Games were held in Athens in 1896, drawing athletes from 14 nations.
Faithfulness score	0.0 — "14 nations" does not appear in the context passage.

4.3 Failure Mode: Comprehension (5% of examples)

One example was faithful to the context but still scored 0.0 for correctness. The model referenced the correct passage and produced a grammatically reasonable answer, but drew the wrong conclusion from the evidence.

Field	Content
Question	What were the two primary consequences of the treaty?
Reference answer	Territorial concessions in the north and a five-year trade embargo.
Model answer	The treaty resulted in territorial concessions in the northern regions.
Correctness score	0.5 — captured one of two required consequences; the trade embargo was omitted.
Faithfulness score	1.0 — all stated claims are supported by the context.

4.4 Strong Performance Example

Field	Content
Question	Who wrote the novel and in what year was it first published?
Reference answer	Mary Shelley, 1818.
Model answer	The novel was written by Mary Shelley and first published in 1818.
Correctness score	1.0 — complete and accurate.
Faithfulness score	1.0 — all claims supported by the context.

5. Limitations

5.1 Dataset Scale

- The evaluation used only 20 examples. This provides limited statistical power: a single additional failure changes the pass rate by 5 percentage points.
- The Dolly 15k dataset was generated by crowd-sourced employees and is not uniformly difficult. Some questions are trivially easy; others require multi-step reasoning. A flat average across all examples obscures this variance.
- A 20-example sample cannot reliably represent the full distribution of closed-QA difficulty. Edge cases and rare failure modes are likely underrepresented.

5.2 Evaluator Reliability

- Both evaluators are LLM-based (GPT-4o-mini, temperature 0). The judge model shares stylistic and knowledge biases with the target model, which may artificially inflate alignment between the two.
- The three-point correctness scale (0.0 / 0.5 / 1.0) is coarse. Borderline cases are subject to judge discretion and may be rated inconsistently across runs.
- No human calibration was performed. LLM judge scores have not been validated against expert human ratings on this specific dataset.

5.3 Model & Configuration Scope

- Only a single model (GPT-4o-mini) was evaluated. Results cannot be generalised to other models without additional experiments.
- Performance is sensitive to the system prompt. A different instruction strategy could yield materially different scores.
- The evaluation was conducted at a fixed point in time. OpenAI model behaviour can change across API versions without notice.

5.4 Task Scope

- Closed-QA is a highly constrained task. These results do not generalise to open-domain QA, summarisation, multi-hop reasoning, or other NLP tasks.

6. Recommendations

6.1 Scale Dataset & Improve Stratification

- Expand to at least 100–200 examples to achieve meaningful statistical confidence. A sample of 20 gives approximately $\pm 13\%$ margin of error at 90% confidence.
- Stratify examples by question type (single-fact, multi-fact, inferential) to enable per-category performance analysis.
- Introduce a 20–30 example “gold” calibration subset with expert human labels to validate LLM judge alignment.

6.2 Improve Target Model Performance

- Prompt engineering: add two or three few-shot examples in the system prompt demonstrating complete multi-part answers to reduce partial-credit failures.
- Chain-of-thought: instruct the model to reason step-by-step before giving its final answer, which may improve performance on inferential questions.
- Explicit anti-hallucination instruction: add the phrase “Do not include any information not explicitly stated in the context” to directly target the 10% faithfulness failure rate.

6.3 Strengthen Evaluation Rigour

- Add a dedicated “completeness” evaluator to separately score how fully the model captures all required facts, decoupling this from the correctness dimension.
- Use a stronger judge model (e.g., GPT-4o) for a calibration run to quantify agreement with the weaker GPT-4o-mini judge.
- Run experiments at temperature > 0 in triplicate and report mean $\pm$ stdev across runs to quantify model variance.

6.4 Operationalise with LangSmith

- Register the best-performing prompt configuration as a named baseline in LangSmith for ongoing regression testing.
- Schedule automated evaluation runs triggered by any prompt or model change to detect regressions before deployment.
- Tag experiments with version identifiers (model name + prompt hash) to enable reproducible comparisons over time and across model upgrades.